

Report
Of
Java Full Stack Development

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: CS-23411121
Name of Student: SHIVANSHU TIWARI
Batch: 2026-27

2026-27

CANDIDATE'S DECLARATION

I, Shivanshu Tiwari, Roll No. CS-23411121, do hereby declare that the internship report titled “Java Full Stack Development” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Shivanshu Tiwari

Roll No.: CS-23411121

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship program. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous mentorship, and the team at EduSkills Academy for structuring and delivering the Java Full Stack internship program that forms the subject of this report.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: Shivanshu Tiwari

Roll No.: CS-23411121

INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE
OF INTERNSHIP







PRESENTED TO

Shivanshu Tiwari

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

Java Full Stack

During the internship, the learner demonstrated proficiency in:

- Introduction to HTML and Web Page Structure
- CSS Fundamentals for Web Design & Bootstrap: Responsive Web Design Framework
- JavaScript: Interactive Web Development & jQuery: Simplifying JavaScript Development
- Core Java: Fundamentals of Java Programming & Advanced Java: Enterprise Application Development
- Spring Framework: Enterprise Java Development
- Spring Boot: Simplifying Java Application Development
- Hibernate: Object-Relational Mapping in Java
- MySQL & Git & Version Control

This certification recognizes the successful completion of the internship requirements.



Issued on: Jun 10, 2026
Certificate ID: 2026-328F130950



Mihir Swain
General Manager
(Learning & Development)
EduSkills

iii

TABLE OF CONTENTS

Candidate's Declaration.....	i
Acknowledgement	ii
Internship Completion Certificate.....	iii
List of Tables.....	v
List of Figures.....	vi
4. Project Description.....	1
4.1 Introduction.....	1
4.2 Organization Profile.....	2
4.3 Problem Statement.....	3
4.4 Project Objectives.....	3
4.5 Scope of the Project.....	4
4.6 Technologies and Tools Used.....	5
4.6.1 HTML, CSS and Bootstrap.....	5
4.6.2 JavaScript and jQuery.....	5
4.6.3 Core and Advanced Java.....	6
4.6.4 Spring Framework and Spring Boot.....	6
4.6.5 Hibernate and MySQL.....	6
4.6.6 Git and Version Control.....	6
4.7 System Architecture.....	7
4.8 Methodology.....	8
4.8.1 Phase 1 — Front-End Foundations.....	9
4.8.2 Phase 2 — Enterprise Java Development.....	9
4.8.3 Phase 3 — Persistence and Collaboration.....	9
4.9 Expected Outcomes.....	9
4.10 Certificates of Completion and Communication Proof.....	10
5. Bibliography / References.....	11

LIST OF TABLES

Table 4.1: Project Scope — In Scope vs. Out of Scope.....	4
Table 4.2: Technologies and Tools Used.....	5
Table 4.3: Week-wise Learning Schedule.....	8

LIST OF FIGURES

Figure 4.1: Layered System Architecture.....	7
Figure 4.2: Request–Response Flow Through the Layered Architecture	8

4. PROJECT DESCRIPTION

4.1 Introduction

The internship is a mandatory, credit-bearing component of the Bachelor of Technology curriculum in Computer Science and Engineering at IILM University, Greater Noida. It is designed to bridge the gap between classroom instruction and industry practice by requiring students to spend a defined period working on structured, guided, real-world-style technical assignments outside the regular academic timetable. This report documents one such internship, undertaken in the domain of Java Full Stack Development, and presents its organizational context, technical content, methodology, and outcomes.

“Full stack development” refers to the ability to design, build, and maintain every layer of a web application — the presentation layer that a user interacts with in the browser, the application and business logic that processes requests and enforces rules, and the persistence layer that stores and retrieves data reliably. A full stack developer is expected to move comfortably between these layers, understanding how a change at one level (for example, a database schema) affects the levels above it (for example, the objects an application manipulates, or the fields a user sees on a form). This breadth of understanding is increasingly sought after in the software industry, where small and mid-sized engineering teams often expect individual developers to own a feature end-to-end rather than specializing narrowly in a single layer.

The specific internship documented here was hosted by EduSkills Academy, the training and skilling arm of the EduSkills Foundation, and was undertaken under the All India Council for Technical Education (AICTE) virtual internship scheme. It was structured as an eight-week (two-month), self-paced, credentialed program in the “Java Full Stack” track, combining guided instructional content with practical exercises across front-end web technologies, core and advanced Java, enterprise application frameworks, and relational database management. The program concluded with the award of an Internship Credential (Certificate ID: 2026-328F130950, issued on June 10, 2026), which is reproduced in the preliminary section of this report.

The remainder of this chapter is organized as follows. Section 4.2 profiles the host organization. Section 4.3 states the problem the internship was intended to address, and Section 4.4 sets out the specific objectives pursued. Section 4.5 defines the scope and

boundaries of the work. Section 4.6 describes, in detail, the technologies and tools covered. Section 4.7 explains the system architecture underlying the applications built during the program. Section 4.8 describes the week-by-week methodology followed, Section 4.9 summarizes the outcomes achieved, and Section 4.10 references the certification and communication proof of completion. Chapter 5 lists the references consulted.

4.2 Organization Profile

EduSkills Academy is the training and skilling division of the EduSkills Foundation, a not-for-profit organization that works to build an “Industry 4.0”-ready digital workforce among engineering and polytechnic students across India. The Foundation's stated mission is to close the gap between academia and industry by giving faculty and students structured access to industry-grade curricula in emerging technology areas such as cloud computing, cybersecurity, networking, automation, and enterprise software development.

EduSkills operates its flagship internship programs in partnership with the All India Council for Technical Education (AICTE) through the AICTE Internship Portal, and co-brands its certificates with established technology companies — among them Cisco, AWS, Red Hat, VMware, Palo Alto Networks, and Juniper — who contribute curriculum content and lend their names to the credentials awarded. The organization reports partnerships with member institutions across more than twenty states in India, delivering programs through a learning-management platform on which students track their own progress, complete modules and assessments, and receive their certificates on completion.

Programs of this kind are offered free of cost to students at AICTE-affiliated or otherwise recognized institutions, run for a fixed duration of two months (eight weeks), and are delivered entirely online in a self-paced format, allowing students to complete modules around their existing academic commitments. Because they are approved under the AICTE virtual internship scheme, many institutions — subject to their own policy — recognize successful completion toward academic internship credit.

The specific offering undertaken for this report is the “Java Full Stack Developer” track. It is positioned by the academy as a response to sustained industry demand for developers who can work across the Java enterprise stack — from front-end interfaces

through to Spring-based back-end services and relational data storage — and is structured across eight weeks covering front-end web technologies, core and advanced Java, enterprise frameworks, and database management, culminating in an Internship Credential upon successful completion of all modules and assessments.

4.3 Problem Statement

Academic curricula in computer science typically teach web fundamentals, programming languages, and database theory as separate courses, each assessed largely in isolation. This is a reasonable way to build foundational understanding, but it leaves comparatively little room for students to practice integrating these separate pieces of knowledge into a single, cohesive, production-style application. As a result, many students graduate with solid theoretical grounding in individual subjects but limited first-hand experience of how those subjects combine in practice — for instance, how a form submitted in a browser eventually becomes a row in a relational database, and everything that has to happen, correctly and in the right order, in between.

This gap is particularly visible in the enterprise Java ecosystem. Concepts such as dependency injection, layered architecture, object-relational mapping, and RESTful service design — all central to how Java is actually used in industry via frameworks such as Spring, Spring Boot, and Hibernate — are rarely covered in depth within a standard academic syllabus, since they sit at the intersection of software engineering practice and language-specific tooling. Similarly, collaborative development practices such as version control with Git are often assumed rather than explicitly taught, leaving students to pick them up informally, if at all, before entering the workforce.

This internship was undertaken specifically to close that gap: to build practical, integrated, full stack development skill by working through a structured curriculum that deliberately connects front-end interfaces, back-end business logic, enterprise frameworks, and persistent storage into a single, coherent application architecture, using the same tools and patterns that are used in industry settings.

4.4 Project Objectives

The internship was guided by the following primary and secondary objectives.

Primary objectives:

- To build end-to-end web applications spanning front-end UI, back-end services, and persistent storage.
- To apply enterprise Java design patterns using Spring, Spring Boot, and Hibernate to develop maintainable, layered applications.
- To gain hands-on exposure to relational database design and collaborative version-control workflows.
- To strengthen core web and Java fundamentals — HTML, CSS, JavaScript, and Core/Advanced Java — that underpin full stack development.

Secondary objectives:

- To develop the discipline of self-paced, structured learning against a fixed timeline and milestone schedule.
- To practise reading and applying official technical documentation independently, as is expected in professional settings.
- To produce a formal, academically structured report suitable for institutional evaluation of the internship.

4.5 Scope of the Project

The scope of the internship covered the complete web development stack relevant to the Java Full Stack track: front-end markup, styling, and responsive design; client-side scripting and DOM interaction; enterprise Java application development using the Spring ecosystem; object-relational mapping through Hibernate; relational database design and querying with MySQL; and collaborative source-control practices using Git. Table 4.1 below summarizes what was explicitly in scope and out of scope for the eight-week program.

Aspect	In Scope	Out of Scope
Front-end	HTML, CSS, Bootstrap, JavaScript, jQuery	Modern SPA frameworks (React, Angular, Vue)
Back-end	Core/Advanced Java, Spring, Spring Boot	Microservices, message queues, GraphQL
Data layer	Hibernate ORM, MySQL	NoSQL databases, data warehousing
Operations	Git & version control	CI/CD pipelines, containerization, cloud deployment

Areas such as cloud deployment, containerization, and microservices architecture were not part of the 8-week curriculum and are noted here as potential directions for further, independent study beyond the scope of this internship.

4.6 Technologies and Tools Used

Table 4.2 summarizes the technologies and tools covered during the internship, grouped by their role in the application stack. Each is described in further detail in the subsections that follow.

Category	Technologies / Tools	Purpose
Markup & Styling	HTML, CSS, Bootstrap	Web page structure and responsive design
Client-side Scripting	JavaScript, jQuery	Interactive, dynamic front-end behaviour
Core Programming	Core Java, Advanced Java	Application logic and enterprise objects
Enterprise Framework	Spring Framework, Spring Boot	Dependency injection, MVC, REST services
Persistence	Hibernate ORM, MySQL	Object-relational mapping and data storage
Collaboration	Git & Version Control	Source-code management and collaboration

4.6.1 HTML, CSS and Bootstrap

The program began with HyperText Markup Language (HTML), covering the structural building blocks of a web page — semantic elements, forms, tables, and the overall document object model that the browser constructs from a page's markup. This was followed by Cascading Style Sheets (CSS), covering selectors, the box model, flexbox-based layout, and typography, and by the Bootstrap framework, which provides a pre-built grid system and component library for producing responsive interfaces that adapt correctly to desktop, tablet, and mobile screen sizes without hand-writing every breakpoint.

4.6.2 JavaScript and jQuery

JavaScript was covered as the language of client-side interactivity — variables, functions, events, and manipulation of the DOM in response to user actions such as

clicks and form submissions. jQuery was introduced alongside it as a library that simplifies common JavaScript operations (DOM traversal, event handling, and AJAX calls) behind a more concise, cross-browser-consistent syntax, which was, at the time the module was written, still in wide use in many existing enterprise front-ends.

4.6.3 Core and Advanced Java

Core Java modules covered the language fundamentals required for enterprise development: object-oriented principles (classes, inheritance, interfaces, polymorphism), collections, exception handling, and I/O. Advanced Java built on this foundation with topics oriented toward enterprise application development, including multithreading basics, generics, and the design of reusable, maintainable business objects intended to be consumed by a service layer.

4.6.4 Spring Framework and Spring Boot

The Spring Framework module covered the core ideas that distinguish enterprise Java development from plain Java programming: inversion of control and dependency injection, the Model-View-Controller (MVC) pattern, and the general shape of a layered Spring application. Spring Boot was then introduced as a convention-over-configuration layer on top of Spring that removes much of the manual configuration traditionally required to bootstrap a Spring application, allowing REST controllers and services to be built and run with minimal setup.

4.6.5 Hibernate and MySQL

Hibernate was covered as the Object-Relational Mapping (ORM) layer that maps Java classes onto relational database tables, removing the need to hand-write most SQL for routine create, read, update, and delete operations. MySQL was used as the underlying relational database, with modules covering schema design, primary and foreign keys, and writing SQL queries directly, so that the mapping performed by Hibernate could be understood rather than treated as a black box.

4.6.6 Git and Version Control

Finally, the program covered Git as the standard tool for source-code version control: initializing repositories, staging and committing changes, branching, and merging. This module was oriented toward the collaborative workflows expected in industry teams, where multiple developers work on the same codebase concurrently and rely on disciplined commit history to coordinate their work safely.

4.7 System Architecture

The applications built during the internship follow a layered architecture, a common enterprise Java design pattern that separates concerns across presentation, application, business, and persistence layers so that each layer can be developed, tested, and modified with minimal impact on the others. Figure 4.1 illustrates this layered structure and the technologies used at each level.

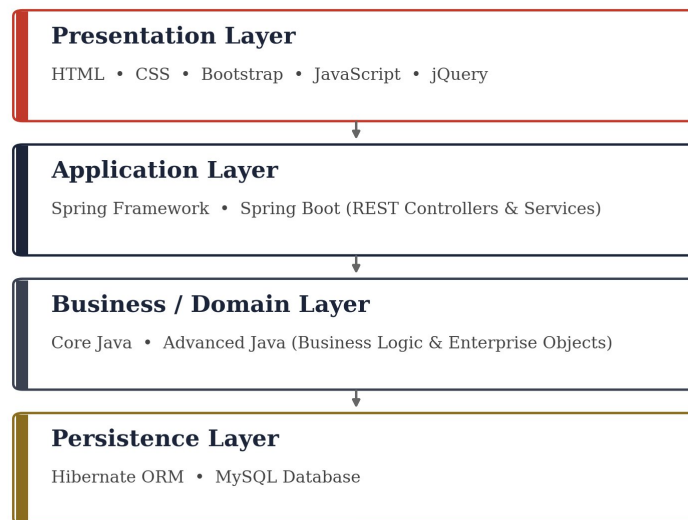


Figure 4.1: Layered System Architecture

The Presentation Layer handles the user interface using HTML, CSS, Bootstrap, JavaScript, and jQuery. The Application Layer, built with the Spring Framework and Spring Boot, exposes REST controllers and services that mediate between the interface and business logic. The Business/Domain Layer, implemented in Core and Advanced Java, encapsulates the application's rules and enterprise objects. Finally, the Persistence Layer uses Hibernate ORM to map Java objects onto a MySQL relational database, providing durable data storage. This separation follows the broader software-engineering principle of separation of concerns: each layer depends only on the one immediately beneath it, and communicates with it through a well defined interface, so that, for example, the database technology could in principle be changed without requiring changes to the presentation layer.

Figure 4.2 traces a single request through this architecture end-to-end, from the moment a user interacts with the browser to the moment a response is rendered back to them. An HTTP request originating in the browser is received by a Spring Boot controller, which delegates the work to the service layer; the service layer applies business rules

and calls a repository, which uses Hibernate to translate the request into SQL executed against the MySQL database. The result then flows back up the same chain — repository to service to controller — before being serialized as JSON or rendered as a view and returned to the browser.

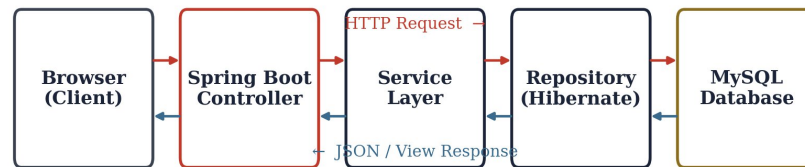


Figure 4.2: Request–Response Flow Through the Layered Architecture

4.8 Methodology

The internship followed a structured, self-paced, module-by-module methodology spread across eight weeks, progressing from foundational front-end concepts to enterprise back-end development and, finally, to persistence and collaborative tooling. Table 4.3 sets out the week-wise schedule that was followed.

Week	Focus Area	Topics Covered
1	Web Foundations	HTML page structure and semantic markup
2	Styling & Layout	CSS fundamentals and the Bootstrap responsive framework
3	Client-side Scripting	JavaScript fundamentals and DOM manipulation; jQuery
4	Java Fundamentals	Core Java: OOP, collections, exception handling
5	Advanced Java	Advanced Java concepts for enterprise application development
6	Enterprise Framework	Spring Framework and Spring Boot for layered, service-oriented apps
7	Persistence	Hibernate ORM and MySQL relational database design
8	Collaboration & Wrap-up	Git & version control; consolidation and assessment

4.8.1 Phase 1 — Front-End Foundations (Weeks 1–3)

The first phase focused on the presentation layer in isolation, without any server-side component. Exercises progressed from static HTML pages, to CSS-styled and Bootstrap-responsive layouts, to pages with JavaScript- and jQuery-driven interactivity such as form validation and dynamic content updates. This phase established the baseline front-end competence on which the later, full stack exercises were built.

4.8.2 Phase 2 — Enterprise Java Development (Weeks 4–6)

The second phase moved to the server side, beginning with Core and Advanced Java to consolidate the language fundamentals, before introducing the Spring Framework and Spring Boot. Exercises in this phase involved designing simple layered applications with controllers, services, and plain in-memory data, so that the architectural pattern could be practiced before persistence was introduced in the following phase.

4.8.3 Phase 3 — Persistence and Collaboration (Weeks 7–8)

The final phase introduced Hibernate and MySQL, replacing the in-memory data used earlier with a real, persistent relational database, and connected the previously separate front-end and back-end exercises into a single, coherent application consistent with the layered architecture described in Section 4.7. The program concluded with a module on Git and version control, and a consolidation period in which the material from all three phases was reviewed ahead of the final assessment that led to the award of the Internship Credential.

Each module across all three phases combined guided instructional content with hands-on exercises, allowing concepts to be applied immediately after introduction, which reinforced retention and practical competence far more effectively than passive reading alone.

4.9 Expected Outcomes

On successful completion of the internship, the following outcomes were achieved, grouped into technical, professional, and personal categories.

Technical outcomes:

- Ability to design and build full stack web applications spanning UI, services, and data storage.
- Working knowledge of enterprise Java development using Spring, Spring Boot, and Hibernate.

- Practical experience with relational database design and querying using MySQL.
- Ability to design and reason about a layered application architecture, as described in Section 4.7.

Professional outcomes:

- Familiarity with collaborative, version-controlled software development workflows using Git.
- Comfort reading and applying official framework documentation independently.
- Experience working to a fixed, self-managed schedule and set of milestones, as summarized in Table 4.3.

Personal outcomes:

- Strengthened problem-solving and debugging skills through iterative, self-paced project execution.
- Increased confidence in independently learning and applying an unfamiliar technology stack.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by EduSkills Academy, confirming successful completion of the 8-week Java Full Stack program (Certificate ID: 2026-328F130950, issued on June 10, 2026), is attached in the preliminary section of this report (see “Internship Completion Certificate,” page iii). The certificate lists the eight core modules completed — spanning HTML, CSS and Bootstrap, JavaScript and jQuery, Core and Advanced Java, the Spring Framework, Spring Boot, Hibernate, and MySQL with Git and version control — consistent with the technologies and methodology described in Sections 4.6 and 4.8 of this report.

Any additional email or platform correspondence confirming enrolment and completion has been retained by the student and can be furnished to the department on request.

5. BIBLIOGRAPHY / REFERENCES

- [1] EduSkills Academy, “Java Full Stack Internship Program,” Course Certificate documentation, 2026.
- [2] EduSkills Foundation, “About Us,” EduSkills Foundation official website. Available: <https://eduskillsfoundation.org/>
- [3] All India Council for Technical Education (AICTE), “AICTE Internship Portal — Java Full Stack Developer Virtual Internship.” Available: <https://internship.aicte-india.org/>
- [4] Oracle Corporation, “The Java Tutorials,” Oracle Java Documentation. Available: <https://docs.oracle.com/javase/tutorial/>
- [5] VMware / Spring Team, “Spring Framework Reference Documentation.” Available: <https://docs.spring.io/spring-framework/reference/>
- [6] VMware / Spring Team, “Spring Boot Reference Documentation.” Available: <https://docs.spring.io/spring-boot/documentation.html>
- [7] Red Hat / Hibernate Team, “Hibernate ORM User Guide.” Available: <https://hibernate.org/orm/documentation/>
- [8] Oracle Corporation, “MySQL 8.0 Reference Manual.” Available: <https://dev.mysql.com/doc/refman/8.0/en/>
- [9] Mozilla Developer Network, “Web Technology Reference: HTML, CSS, JavaScript.” Available: <https://developer.mozilla.org/>
- [10] Bootstrap Team, “Bootstrap Documentation.” Available: <https://getbootstrap.com/docs/>
- [11] Git, “Pro Git Book,” Available: <https://git-scm.com/book/en/v2>